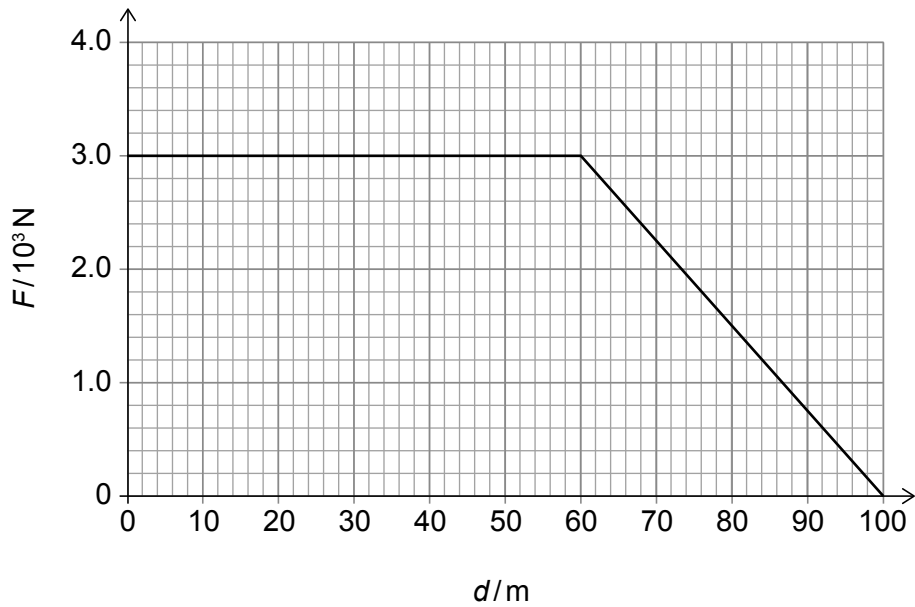


18. (a) A car of mass 1600 kg accelerates from rest.

The graph shows how the resultant force F acting in the direction of motion of the car varies with the distance d travelled by the car.



(i) State what is represented by the area under the graph. [1]

(ii) Calculate the final speed of the car. [2]

(This question continues on the following page)

PAPER 2 / QUESTION 18

2225-6709 / original question 1

(Question 18 continued)

A different car travels on a horizontal road at a constant speed of 45 m s^{-1} . The engine of the car develops a power of 140 kW . The resistive force F_d acting on the car is given by

$$F_d = cv^2$$

where v is the speed of the car and c is a constant.

(b) Determine c . State the fundamental SI unit for your answer.

[3]

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